Experiment 21: Syntax-Driven Semantic Analysis in Python

1. Aim

To write a Python program that performs syntax-driven semantic analysis by extracting noun phrases and their meanings from a sentence.

2. Procedure

1. Import `nltk` and `wordnet` in Python.
2. Tokenize and POS-tag a sample input sentence.
3. Use `nltk.RegexpParser()` with grammar `NP: {<DT>?<JJ>*<NN>}` to extract noun phrases.
4. Retrieve the head noun of each extracted `NP` chunk and look up its WordNet meaning.
5. Display the extracted noun phrases and their semantic definitions.

3. Program

```
import nltk
from nltk.corpus import wordnet as wn

tagged = nltk.pos_tag(nltk.word_tokenize("The smart boy read a book"))
tree = nltk.RegexpParser("NP: {<DT>?<JJ>*<NN>}").parse(tagged)

for sub in tree.subtrees(lambda t: t.label() == "NP"):
    words = [w for w, t in sub.leaves()]
    print("NP:", " ".join(words), "| Meaning:", wn.synsets(words[-1])[0].definition())
```

4. Output

```
NP: The smart boy | Meaning: a youthful male person
NP: a book | Meaning: a written work or composition that has been
published
```

5. Result

Thus, the Python program was successfully executed to perform syntax-driven semantic analysis by extracting noun phrases and their meanings.